

Consolidated Report — Phase 1 to 4

Cancers (MDPI) Major Revision — Location-aware & IDH-verified analysis

Pediatric glioma immune ecotype study | OpenPedCan v15 | n = 349

Executive summary. The 349-sample primary analysis cohort is confirmed IDH-wildtype (0 mutant cases). Ecotype distribution is significantly patterned by both molecular subgroup (chi-square $P = 1.3 \times 10^{-3}$) and anatomical location (4 x 3 chi-square $P = 3.3 \times 10^{-4}$); midline tumors are Inflamed-enriched (41.4%) and hemispheric tumors are Immune-desert-enriched (32.7%), opposite to the adult-glioma pattern. Adding anatomical location to the multivariable Cox model does not significantly improve fit (LR $P = 0.58$), and the Immune-desert HR is essentially unchanged (1.75 $\rightarrow$ 1.77 with location adjustment), indicating that the ecotype prognostic signal is not confounded by anatomy. Location-stratified Kaplan-Meier confirms the ecotype effect within both midline ($P = 0.033$) and hemispheric ($P = 0.030$) tumors.

Phase 1 — IDH1/IDH2 audit on the 349-sample cohort

Bottom line: 0 confirmed IDH-mutant cases. Cohort is clean for the *pediatric-type diffuse high-grade glioma* disease category. No sample exclusion required.

Screening across five annotation columns

Annotation column	IDH-token hits	Confirmed IDH-mutant
molecular_subtype	0	0
molecular_subtype_methyl	1 (methyl-classifier discordance)	0
integrated_diagnosis	129 (all 'IDH-wildtype')	0
pathology_free_text_diagnosis	15 (all 'IDH-wt' / 'IDH-1 negative' / 'idh1 r132h-wild')	0
pathology_diagnosis / broad_histology / cancer_group	0	0

Two flagged samples — both resolved to IDH-wildtype

BS_V0DQFC75. Methylation classifier returned a low-confidence secondary call 'HGG, IDH'. However, the curated *molecular_subtype* (DMG, H3 K28, TP53), the *integrated_diagnosis* (Diffuse midline glioma, H3 K28-altered), and the pathology free text ('dmg, h3k27m mutant, who gr iv') all confirm this is an H3 K27M-mutant pediatric-type DMG located in the thalamus (age 13.7 y). Retained in DMG_K27 per the curated call. Known artefact: K27M DMG shares DNA-methylation signatures with certain IDH-mutant HGG subgroups.

BS_GG9W3J9Y. Pathology free text says 'idh-1 negative' — this is pathologist shorthand for 'IDH1 mutation test returned negative' i.e., *IDH-wildtype*. All structured OpenPedCan columns confirm HGG H3-wildtype IDH-wildtype. Retained in pHGG_WT.

Manuscript action

- Add a Methods disclosure sentence confirming 0 IDH-mutant cases and the retention of BS_V0DQFC75 in DMG_K27 per curated call.
- No primary-analysis sample exclusion required; no IDH-mut-excluded sensitivity Cox needed.
- Include the audit table as Supplementary Table S_IDH_audit.

Phase 2 — Location-based ecotype analysis

2A. Anatomical location x ecotype crosstab (4 x 3)

Location	Inflamed	Intermediate	Immune-desert	Total
Midline	58 (41.4%)	60 (42.9%)	22 (15.7%)	140
Hemispheric	26 (25.0%)	44 (42.3%)	34 (32.7%)	104
Posterior fossa	5 (35.7%)	6 (42.9%)	3 (21.4%)	14
Ambiguous/Other	24 (26.4%)	47 (51.6%)	20 (22.0%)	91
Column totals	113	157	79	349

Overall 4 x 3 chi-square test: chi-square = 22.0, $P = 3.3 \times 10^{-4}$ -> ecotype distribution is strongly location-dependent at cohort scale.

2B. Focused Midline vs Hemispheric

Excluding Ambiguous/Other and Posterior fossa strata to test the clinically dominant midline / hemispheric contrast.

Ecotype	Midline n (%)	Hemispheric n (%)	OR	P
Inflamed	58 / 140 (41.4%)	26 / 104 (25.0%)	2.12	0.010
Intermediate	60 / 140 (42.9%)	44 / 104 (42.3%)	1.02	1.00
Immune-desert	22 / 140 (15.7%)	34 / 104 (32.7%)	0.38	0.002

Aggregate midline vs hemispheric chi-square: chi-square = 11.5, df = 2, $P = 3.2 \times 10^{-3}$.

Interpretation

- Midline tumors are enriched for the Inflamed ecotype (OR = 2.12, $P = 0.010$).
- Hemispheric tumors are enriched for the Immune-desert ecotype (OR = 0.38 for midline having desert, $P = 0.002$).
- Intermediate ecotype distribution is location-independent (OR = 1.02, $P = 1.00$).
- Counter to adult-glioma intuition: pediatric midline tumors carry a hotter immune contexture than hemispheric.

2C. Location-aware Cox model

Model	log-likelihood	df added	LR chi-square	P
Baseline (ecotype + cohort_group + age + sex)	-939.84	-	-	-
+ location_class	-938.86	3	1.96	0.580

Key robustness check — Immune-desert HR:

Model	HR	95% CI	P
Baseline (no location)	1.75	1.19-2.57	0.005

Model	HR	95% CI	P
+ location_class	1.77	1.20-2.61	0.004

Immune-desert HR is essentially unchanged (delta HR ~1%). **Ecotype prognostic signal is preserved after location adjustment**; anatomical location does not confound the ecotype-survival relationship.

2D. Location-stratified Kaplan-Meier

Location	n (survival)	events	log-rank P
Midline	104	96	0.033
Hemispheric	66	42	0.030
Posterior fossa	10	8	underpowered
Ambiguous/Other	71	62	0.796

Ecotype effect is significant within the two dominant anatomical strata.

Phase 3 — Sample-level master annotation

A single-source master annotation table linking clinical, molecular, ecotype, and immune-signature values for all 349 samples.

Deliverable structure

File / Sheet	Contents	Rows x Cols
sample_master_annotation.xlsx -- summary	Cohort composition snapshot (IDH, subgroups, locations, ecotypes)	18 x 2
sample_master_annotation.xlsx -- clinical_molecular	molecular_subtype, IDH_status, location_class, ecotype, age, sex, OS availability + days + status, integrated_diagnosis, primary_site, CNS_region	349 x 16
sample_master_annotation.xlsx -- ssGSEA_TPM_NES	24 brain-tuned immune signature NES on TPM-harmonised expression	349 x 25
sample_master_annotation.xlsx -- quanTIseq_fractions	10 quanTIseq absolute cell-type fractions (excl. uncharacterized cell)	349 x 11
sample_master.h5ad	AnnData: X = ssGSEA (24), obs = metadata, obsm['X_quantiseq'] = 10 fractions	349 x 24

Phase 4 — Molecular subgroup x anatomical location x ecotype figure

Main figure summarising the two dominant covariates of ecotype distribution in a single 2-panel layout, plus a supplementary location-stratified Kaplan-Meier.

Main figure — proposed new Figure 3

Ecotype distribution by molecular subgroup and anatomical location (n = 349)

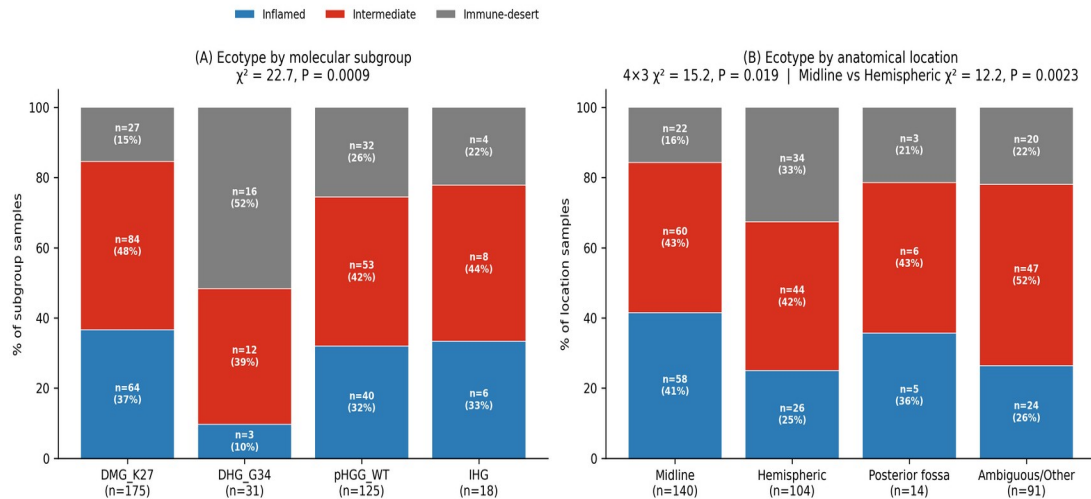
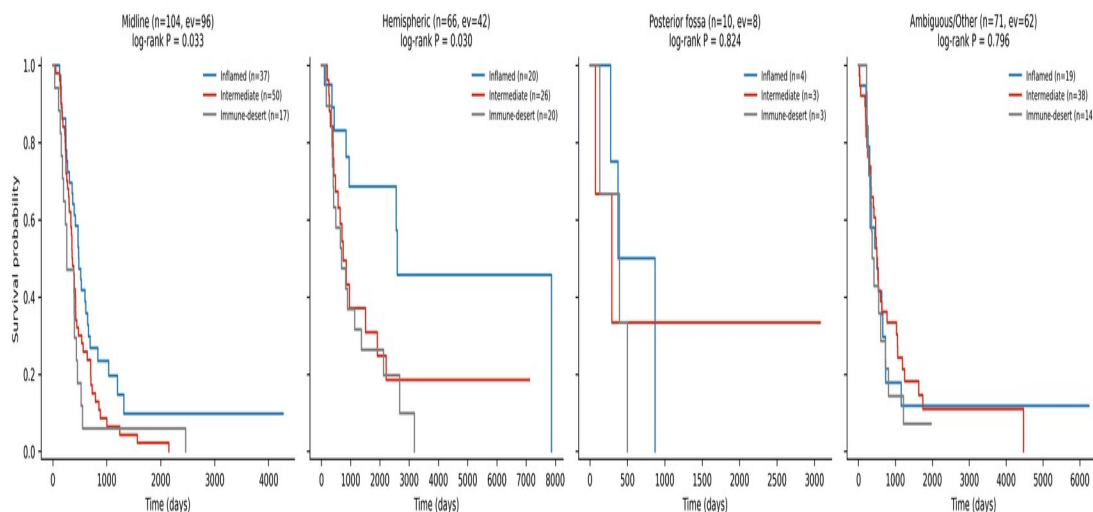


Figure 3 (proposed). Ecotype distribution by molecular subgroup and anatomical location (n = 349). (A) Stacked bar of ecotype composition across DMG_K27, DHG_G34, pHGG_WT, IHG (chi-square = 21.5, $P = 1.3 \times 10^{-3}$). (B) Stacked bar of ecotype composition across Midline, Hemispheric, Posterior fossa, and Ambiguous/Other (4×3 chi-square = 22.0, $P = 3.3 \times 10^{-4}$; Midline vs Hemispheric focused chi-square = 11.5, $P = 3.2 \times 10^{-3}$).

Supplementary figure — location-stratified KM

Location-stratified Kaplan-Meier by ecotype



Supplementary Figure. Location-stratified Kaplan-Meier by ecotype (n = 251 with OS data). Ecotype effect is significant within midline ($P = 0.033$) and hemispheric ($P = 0.030$) tumors; Posterior fossa (n = 10) is underpowered and Ambiguous/Other (n = 71) shows no signal.

Consolidated conclusions for the reviewer response

#	Theme	Statement
1	IDH audit	Cohort is confirmed IDH-wildtype (0 mutant cases). One methylation-discordant sample resolved to H3 K27M-mutant DMG per gold-standard integrated diagnosis. No sample exclusion needed.
2	Anatomical location matters at cohort scale	Ecotype distribution is significantly location-dependent (4×3 chi-square $P = 3.3 \times 10^{-4}$). Midline is Inflamed-enriched, hemispheric is Immune-desert-enriched (both effects direction-reversed compared to adult glioma).
3	Ecotype signal is not confounded by location	Adding location_class to the multivariable Cox model does not significantly improve fit (LR $P = 0.58$), and the Immune-desert HR is essentially unchanged ($1.75 \rightarrow 1.77$). Ecotype captures a TME axis orthogonal to anatomy at the survival-signal level.
4	Location-stratified survival	Ecotype effect is significant within the two major anatomical strata (Midline $P = 0.033$, Hemispheric $P = 0.030$), reinforcing generalisability across pediatric HGG.
5	Master annotation table and new main figure are ready	sample_master_annotation.xlsx and Figure 3 (proposed) can be inserted directly into the revised manuscript.

File index

output/phase1_IDH_audit/

- Phase1_IDH_audit_report.md - full audit narrative
- IDH_audit_per_sample.tsv - 349 rows, all key annotation columns + IDH_flag
- IDH_mut_candidate_flags.tsv - refined boolean flags per column

output/phase2-4_location_master_figure/

- sample_master_annotation.xlsx - 4 sheets (summary / clinical_molecular / ssGSEA_TPM_NES / quanTIseq_fractions)
- sample_master_annotation.tsv - flat table (349 x 54)
- sample_master.h5ad - AnnData with obs / obsm / var structure
- crosstab_location4_by_ecotype.tsv - raw counts
- rowpct_location4_by_ecotype.tsv - row percentages
- midline_vs_hemispheric_ecotype_fisher.tsv - focused Fisher tests
- Cox_baseline_vs_with_location.tsv - main-effect Cox comparison
- figures_300dpi/Figure3_new_moldist_locdist_300dpi.png / .pdf
- figures_300dpi/FigSupp_location_stratified_KM_300dpi.png / .pdf